

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Specification for Mainline Fittings
Document Number	EA-BFPCL-GPMC-GGPL-FEED- PPL-SPC-004
Document Revision	R01
Document Status	Issued for Review
Originator	Emeka Nwachukwu
Security Classification	Restricted
Issue Date	04-Aug-22

Specification for Mainline Fittings

R01	04.08.22	Issued for Review	E. Nwachukwu	M.Ekpunobi	K. Yusuff	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	04.08.22	Issued for Review

Revision Philosophy:

- All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED- PPL-SPC-004

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002	Wall Thickness Calculation Report
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-001	Specification for Line Pipe
EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-006	Data Sheets for Induction Bend
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-010	Pipeline Cleaning, Gauging, & Hydrotesting Specification.
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
ASME	American Society of Mechanical Engineers
API	American Petroleum Institute
CS	Carbon Steel
DEP	Design and Engineering Practices
DP	Design Pressure
FEED	Front End Engineering
ISO	International Organisation for Standardization
3LPE	Three-Layer Polyethylene
MMscf/d	Million standard cubic feet per day
NDE	Non-Destructive Examination
MPS.	Manufacturing Procedure Specification
NPS	Nominal Pipe Size
OD	Outer Diameter
PSL	Product Specification Level
TD	Design Temperature
SMTS	Specified Minimum Tensile Strength
SMYS	Specified Minimum Yield Strength
WT	Wall thickness
WPS	Welding Procedure Specification
WPQR	Welding Procedure Qualification Record

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

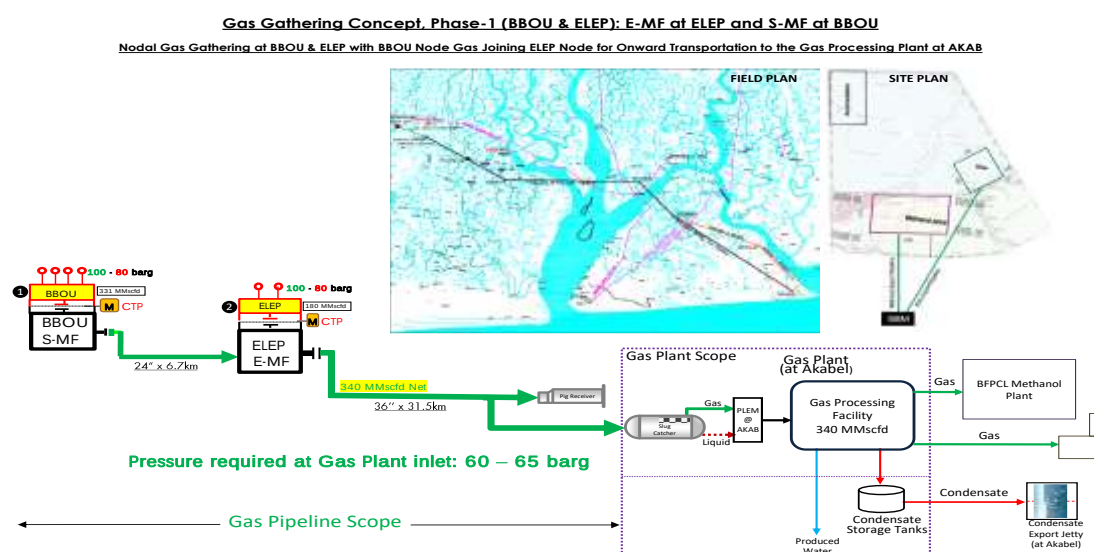


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification defines the technical requirements for the design, materials, manufacture, testing and inspection of forged carbon steel flanges of the welding neck type and branch outlet fittings to be used in the pipeline systems of BFPCL Gas Gathering Pipeline Project Phase 1.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 31.40.00.10-Gen	Pipeline Engineering (Amendments/Supplements To ISO 13623)
DEP. 82.00.10.10-Gen	Project Quality Assurance

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Environmental and Pipeline Design Data

3.1. Environmental Data

Refer for the basis of design document EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002

3.2. 24" Multiphase Pipeline Data

The 24" pipeline design data for the Multiphase Pipeline Project are as shown below

Parameter	Description
Design Code	ASME B31.8
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Design Life	30 years
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L-X70(PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength	483 Mpa
NPS	24"
Ultimate Tensile Strength	570 MPa
OD / Wall Thickness	15.88mm
Design Factor	0.6
Corrosion Allowance	3 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Cathodic Protection	Impressed current

3.3. 36" Multiphase Pipeline Design Data

The 36" pipeline design data for the Multiphase Pipeline Project are as shown below

PARAMETER		DESCRIPTION
From	ELEPA Manifold Station	
To	AKABEL Station	
Pipeline Length	31.5km	
Design Life	30 years	
Process Fluid	Condensate/hydrated Hydrocarbon Gas.	
Design Pressure	110 barg	
Operating Pressure	100 barg	
Max Design Temperature	60°C	
Min Design Temp (Pipe-	12.7°C	
Min Design Temperature	-13.3°C	
Line Pipe Material Grade	API 5L-X70 (PLS2) SAWL - 100% radiographed.	
Installation Temperature	23°C	
Specified Minimum Yield	483 Mpa	
NPS	36"	
Ultimate Tensile Strength	570MPa	
Wall Thickness	23.83mm	
Design Factor	0.6	
Corrosion Allowance	6 mm	
ANSI Class	900#	
Parent Pipe External Coat-	3LPE	
Cathodic Protection	Impressed current	

4. General

4.1. Definition

Throughout this specification, the following definitions apply:

Purchaser;

Means the Company or the Supplier (as applicable), which have placed the Purchase Order to the Manufacturer.

Inspector;

Means the Company's representative(s) of member(s) from an Inspection Agency duly appointed by the Company or the Supplier (as applicable) to act as its representative(s) for the purpose of the Contract.

Approval;

Means the authorisation in writing given by the Company to the Supplier to proceed with the performance of a specific part of the work without releasing in any way the Supplier from any of its obligations or liabilities under the Contract or at law; Approve, Approved and Approval shall be construed accordingly.

4.2. Approval of Supplier's procedures

The Supplier shall submit the following documentation for review and approval one month prior to commencement of fabrication.

- Design calculations and fabrication drawing of flanges.
- Design calculation of gaskets and bolts
- Manufacturing Procedure Specification, describing all proposed forging operations, heat treatment, inspection and non-destructive testing in their correct sequence. This document will also mention the characteristics of equipment used to carry out above mentioned operations

Sketch showing rough forging piece before machining and, when applicable, the test blank with exact location and orientation of test specimens

Heat treatment and recording procedures, including number and location of thermocouples

- Dimensional inspection procedures
- Non-destructive inspection procedures
- Quality Control Plan
- Welding Procedures Specifications (where applicable)
- Welder qualifications (when applicable)
- Handling, storage and packing procedures
- Proposed measures for temporary corrosion prevention (non-coated flanges).

Once approved these procedures shall not be altered without the written approval of the Company.

The Supplier shall also submit for approval any deviations to requirements specified in the other applicable technical documents referenced herein.

4.3. Quality Assurance / Quality Control

The Supplier shall operate a Quality Assurance system in accordance with ISO TS 29001, approved by a recognised authority. The Quality Assurance Manual shall be submitted prior to the start of work.

Four weeks prior to the commencement of the flange manufacture, the Supplier shall prepare and submit a quality Control Plan which describes the inspections to be performed during the supply of flanges. The Quality Control Plan shall set forth "witness", "hold", "review" and "monitor" points.

5. Design

5.1. Design Data

Details of pipe diameter, wall thickness and material grades of the matching pipelines shall be stated in flange Data Sheet to be presented as per the form shown in Appendix 1 of the present specification, together with the relevant design conditions.

5.2. Flange Design

- The Supplier shall provide design calculations and fabrication drawings in accordance to MSS SP 44 (flanges) or MSS SP 97 (branch outlet fittings) and ASME section VIII, Division 1 Appendix 2, for approval. Specially, the flanges shall be checked / designed for the operating and extreme conditions. Equivalent pressure due to bending moment in the flange shall be considered, as well as decrease of allowable stresses due to flange service temperature.
- However, the design factor for hoop stress, longitudinal stress and combined stress shall comply with ASME B31.4 or ASME B31.8, as applicable, and shall be, as a minimum, that of the matching pipeline (specially at flange hub).
- The dimensions of flanges shall be, as a minimum, those specified in ASME B16.5 for NPS ½" through 24", and ASME B16.47 for NPS 26" through 60".
- Design calculations shall also be provided for flange bolts (pre-tensioning included) and ring joint for the intended service conditions.
- The required flange facing (i.e. raised face, ring type joint) shall be clarified in the data sheets.
- Flange facings (including ring joint grooves) shall be in accordance with MSS SP 44 and ANSI B 16.5, as applicable.
- All flanges shall be designed with material tensile properties in accordance with requirements of MSS SP 44 and at least equal to those of the matching pipes for the welding neck (refer to 5.4 of this specification). If lower yield strength is proposed in the ring section (plateau), this proposal shall be assessed by calculation note for Company approval.
- The nominal internal diameter (bore) shall be that of the matching pipeline as defined in the data sheet with a tolerance of ± 1.6 mm.
- The minimum wall thickness at weld neck (or hub) shall not be less than that of the matching pipeline. The flange hub shall be designed so that it can be easily fitted to the matching pipeline.

6. Flange material

6.1. Specification

- Flange material shall comply with ASTM A 694 and the requirements stated in the following.

6.2. Manufacturing Process

- Steel shall be made by the Electric Furnace or Basic Oxygen Converter, and shall be fully killed and made to fine grain practice.
- The material shall be hot forged to bring it as close as possible to the final shape and size of flange. All parts shall undergo a reduction ratio of at least 2:1 during the forging process.
- All flanges shall be normalised or quenched and tempered after forging. Tempering shall take place one hour after quenching.

6.3. Chemical Composition

- Unless alternative compositions are proposed by the Supplier and approved at the bid stage, the chemical analysis shall comply with the following for ladle and product.

Table 5.3: Material Composition

Element	Weight % max	
	Ladle	Product
Carbon	0.16	0.18
Manganese	1.50	1.60
Silicon	0.35	0.40
Aluminium (total)	0.060	0.060
Phosphorus	0.025	0.025
Sulphur	0.025	0.025
Niobium	0.06	0.06
Vanadium	0.06	0.06
Nickel	0.80	0.80
Chromium	0.25	0.25
Molybdenum	0.30	0.30
Copper	0.30	0.30
Cr + Mo + Cu	0.50	0.50
Nitrogen	0.015	0.015

- Aluminium to nitrogen ratio shall be at least 2.
- No alloying additions of elements other than those listed above shall be made without prior approval of the Company.
- The Supplier shall perform analysis for all the elements listed above and any other intentionally added elements.
- The Supplier shall carry out a chemical analysis on each heat of steel used and also on

two products per heat. The maximum Carbon Equivalent shall be 0.42% on ladle analysis and 0.43% on product analysis, when calculated with the following formula:

$$CE = C + \frac{Mn}{6} + \frac{Cr+Mo+V}{5} + \frac{Cu+Ni}{15}$$

- If any part of the product analyses fails to meet the requirements stated above, the whole heat shall be rejected or each individual product shall be fully analysed and all products failing to meet the requirements shall be rejected.
- For flange diameter above 16", chemical analysis shall be carried out on every product.

6.4. Mechanical Properties

6.4.1. Sampling and Pequency of Testing

- The Supplier shall test specimens cut from a complete production forging from locations and with orientations shown in Appendix 2 and Appendix 3.
- For flange 24" and larger, testing on separately forged test blanks may be made provided that shape and size of test blanks (to be proposed by the Supplier) are quite realistic against size of the products to test and that approval has been obtained from the Company at the bid stage.
- The sampling requirement shall be one full set of tensile and Charpy tests per each steel heat, flange size and heat treatment batch.
- Grouping of flange sizes and materials may be agreed with the Company at the bid stage according to manufacturing process.
- Hardness tests shall be carried out on all flanges.

6.4.2. Tensile Properties

- Tensile tests shall be carried out in accordance with ASTM A 370. The largest practical test specimen size shall be used. The required tests results are stated in the following table.

Table 5.4.2 Material Strength Properties

Mechanical properties	Flange material grade (ASTM A 694)						
	F 42	F 46	F 52	F 56	F 60	F 65	F 70
Minimum Yield Strength (0.2% proof stress) (MPa)	290	317	358	386	413	448	482
Minimum UTS (MPa)	413	434	455	468	517	530	565
Minimum Elongation in 2 inches (%)	23	23	22	21	20	20	18
Max YS/UTS	≤ 0.85	≤ 0.85	≤ 0.85	≤ 0.85	≤ 0.88	≤ 0.90	≤ 0.90
BRINELL Hardness (HB)			136-198	139-190	155-225	158-210	165-220

- For NPS 38 and larger sizes of the 300 and higher classes of welding neck flanges shall have 42 ksi minimum yield strength in the ring section (plateau), unless otherwise specified into the Data sheet

6.4.3. Impact Properties

- Charpy V-notch tests shall be carried out in accordance with ASTM A 370 and ASTM E 23 at locations shown in sketches of Appendix D-1 of MSS SP44. Each impact test shall consist of three specimens. Full size specimens (10 mm x 10 mm) shall be used.
- The minimum required impact energies are stated in the table below. The impact test temperature shall be stated in the Data Sheet.

Table 5.4.3: Charpy V-notch Impact

Grade	Minimum Average Value	Minimum Individual Value	Minimum Shear area
F 42	34	26	85% (*75%)
F 46	36	27	85% (*75%)
F 52	40	30	85% (*75%)
F 56	44	33	85% (*75%)
F 60	48	36	85% (*75%)
F 65	52	39	85% (*75%)
F 70	56	42	85% (*75%)
Note: Lateral expansion of broken specimen shall be recorded.			

6.4.4. Hardness Survey

- A hardness survey shall be carried out on each produced flange after final heat treatment, in accordance with ASTM E 10, or equivalent. A minimum of eight equi-spaced readings shall be carried out on the outside edge close to the welding neck (i.e. on flange "hub"). A further minimum of eight readings shall also be made at random locations on the thicker part of each flange (i.e. on flange "ring").
- Hardness shall not exceed the hardness range shown in table 5.4.2 above, at any point. The Inspector reserves the right to ask for hardness checks at any other location on the flange, e.g. when any doubt about consistency of results occurs.
- Hardness testing shall be performed using portable equipment approved by the Company.

6.4.5. Simulated PWHT

- When so required and stated in the data sheet (and at least when wall thickness at flange welding end exceeds 31.8 mm), tensile tests and impact tests as per 5.4.2 and 5.4.3 above shall be repeated after a simulated Post Weld Heat Treatment (PWHT) has been made on the flange material at a
- holding temperature of $620^{\circ}\text{C} \pm 10^{\circ}\text{C}$ for one hour per 25 mm thickness of flange hub. The tests and test results shall meet the requirements of this specification.

- PWHT tests are required on flanges when a PWHT is made by the Manufacturer on weld between flange and pup piece, if any. This happens as a minimum when wall thickness of matching pipe exceeds 31.8 mm.

6.4.6. Failed Tests

- The Inspector will only permit a retest if there is reason to believe that failure was due to some fault in preparation or testing and that the result was not representative of the test sample. Otherwise, the failed test shall be carried out on every flange in the heat or lot, or alternatively, the entire heat or lot shall be re-heat treated and the full set of tests carried out on one flange.

6.4.7. Storage of Test Specimen

- The test specimen and the remaining part of tested pipes shall be stored by Manufacturer till the final release of goods. Those pieces may be used for any further investigation or counter tests as addressed in that present specification.

6.4.8. Material Rings for Welding Tests

- When it is agreed that flanges can be supplied without welded pup pieces, then the Supplier shall supply three (3) test rings to the Company for qualification of weld procedures to be made by others. The rings shall be of the same material and size of the hub and also shall have been forged and heat treated in the same way as the related supplied flanges.
- Test rings shall have a minimum length of 300 mm. Three rings are required for each size and material grade of flanges. Both ends of the rings shall be bevelled as for the related bends of the order.
- Marking of test rings shall be agreed with the Company.

7. Welding Necks Ends, Gaskets and Bolting Materials

7.1. Welding Necks Ends

- The flange welding neck bevel shall be as per MSS SP 44, except that root face shall be $1.6 \text{ mm} \pm 0.6 \text{ mm}$.

7.2. Gaskets

- The gasket material shall be complying with MSS SP 44 and shall be selected in agreement with the Company.
- Ring joint gaskets shall be made to ANSI B 16.20 and MSS SP 44 and shall be manufactured from soft iron.

7.3. Bolting

- Bolting shall be in accordance with MSS SP 44. The data sheet may state the required specifications and grades for bolts and nuts, where necessary. Surface treatment is not recommended. After assembly, stud bolts and nuts shall be cleaned (washed and degreased) abrasive blasted and painted as per project specification
- The bolts provided shall allow for use hydraulic tightening system.

7.4. Electrical Continuity

The Supplier shall ensure that the flanges shall achieve the electrical continuity of the pipeline after installation. The bolting shall be in electrical contact with the flanges

8. Blind Flanges

When required, blind flanges with a 1"½ valve and plug, suitable for hydrotest, will be supplied with the flanges. Blind flanges will be forged.

- These items are temporary items and bolts and gaskets used for the connections to the flange are also considered as temporary items (i.e. they will not be re-used for the final connection).
- These temporary items have to be designed (flange rating) by the Supplier for one year design life on the construction site.
- Blind flanges and gaskets shall comply with MSS SP 44 and ANSI B 16.20.

9. Fittings

The material shall be hot forged to bring it as close as possible to the final shape and size of flange. All parts shall undergo a reduction ratio of a least 2:1 during the forging process. Branch outlet fittings shall be manufactured in accordance with MSS SP 97. The sizes and wall thicknesses will be indicated in Purchase Requisition.

All flanges and branch outlet fittings shall be normalized or quenched and tempered after forging. Tempering shall take place one hour after quenching.

10. Inspection and non-destructive testing

10.1. General

- The Inspector shall have access to the workshops at all times when work is in progress and shall be at liberty to inspect the manufactured items at any stage and to reject any part not complying with this specification.
- Each part of the work shall be inspected by the Supplier during manufacture. Sufficient inspections shall be made by the Supplier to ensure that the materials, manufactured items and testing comply in all respects with the requirements of this specification.
- The Company inspection shall not relieve the Supplier from its responsibility, being understood that the Supplier shall remain responsible for quality and inspections.

10.2. Non-Destructive Testing (NDT)

10.2.1. General

- The Supplier shall submit detailed inspection and NDT procedures for review and approval. This shall also detail the full extent of the inspection.
- The Supplier shall establish, before commencing production, an identification system such that each flange can be fully identified against the Mill test certificates.
- The qualifications of the NDT personnel used by the Supplier shall be subject to approval by the Inspector. The Inspector reserves the right to retest and reject any of the NDT personnel at any time.

10.2.2. Extent of Inspection

Flanges shall be subject to the following inspection:

- 100% visual inspection of internal and external surfaces
- 100% magnetic particle inspection of internal and external surfaces
- 100% ultrasonic inspection.

Final NDT shall be carried out after heat treatment with the forging in the finish-machined condition

10.2.3. Visual Inspection

Forgings shall comply with the workmanship, finish and appearance requirements of ASTM A 694.

10.2.4. Magnetic Particle Inspection (MPI)

MPI shall be carried out over the entire surface area of the forgings in accordance with SE 709, Article 25 of ASME V, using wet particles and the yoke method. No linear indication shall be permitted.

10.2.5. Ultrasonic Inspection

Ultrasonic testing shall be carried out in accordance with ASME V or ASTM A 388/A 388M.

10.2.6. Straight Beam Examination

Indications which are equal to or exceed that obtained from a 5 mm diameter flat bottom hole located at mid-thickness in a calibration test piece of thickness equal to half the thickness of the work piece or 50 mm, whichever is greater, are unacceptable.

10.2.7. Angle Beam Examination

Indications which are equal to or exceed those obtained from a 60 deg. V-notch 25 mm long and having a depth no greater than 5% of the nominal wall thickness in a test piece, are unacceptable.

11. Repair of Defects

Minor surface defects on flanges shall be ground out leaving a smooth gradual profile provided that the wall thickness is not reduced below the nominal value given in MSS SP 44 or by proper calculations for non-standard flanges. Any ground areas (except for cosmetic grinding) shall be examined by Magnetic Particle Inspection. Forgings containing larger defects shall be rejected. Repair welding of forgings is forbidden.

12. Dimensional Requirements and Inspection

Flanges shall be machined according to the dimensions and tolerances of MSS SP 44 for sizes above NPS 10", and of ANSI B 16.5 for NPS 10" and smaller, or as otherwise agreed between the Company and the Supplier for non-standard flanges.

In all cases, the nominal internal diameter of the flange shall be the same as the nominal internal diameter of the matching pipeline as shown on the data sheet, with a tolerance of ± 1.6 mm

This shall be checked in the presence of the Inspector using internal callipers, gauges or equivalent approved devices.

The finish of contact faces of flanges shall be checked on each flange in full compliance with MSS SP 44.

Profile and dimensions of weld neck ends are defined in section 6 of this specification. This requirement also applies to welded pup pieces, if any.

The dimensional inspection shall be carried out after welding the pup piece (if any) to the flange.

13. Preparation for Shipment

13.1. Mill Hydrostatic Testing

The Supplier shall certify that the flange (including pup pieces, if any) can withstand the hydrostatic test pressures of 90% of SMYS of matching pipe (using the Barlow's formula and the minimum specified wall thickness of matching pipe for calculation).

13.2. Protection

The bevelled ends and contact faces of flanges shall be protected by a peel-off varnish. When required in the data sheet, flanges (or their pup pieces) shall be fitted with bevel protectors of a type approved by the Company.

13.3. Marking of Flanges

Marking shall be such that complete traceability is demonstrated in all stages of manufacture. Each flange shall be clearly identified on the outside rim using the following marking:

- Company's name
- Project name
- Purchase Order no.
- Mill Name / Mark
- Steel grade
- Item No. / Requisition No.
- Heat number
- Pressure-temperature rating (MSS SP 44)
- Nominal pipe diameter
- Weight (in kg).

Marking shall be approved by the Inspector prior to the flange production. The original marking of the matching pipes shall be transferred to each corresponding pup piece or scrap of pipe.

The flange number and class rating shall be stamped on the periphery of each flange ring. The mark of the Inspector shall also be added.

13.4. Handling, Storage and Packing

The Supplier shall ensure that all finished flanges are handled and stored in such a manner so as to preserve the machined faces and general surface condition of the forging.

The Supplier shall ensure that all flanges are adequately protected against corrosion on a temporary basis (until installation).

The flanges shall be stored clear of the ground and adequately protected from water and mud.

The Supplier shall take suitable corrective action if damage is observed by the Inspector due to inadequate handling or storage conditions.

13.5. Documentation

Mill certificates 3.1.C according to EN 10204 Std are required for the flanges. Mill certificates shall be issued by the Manufacturer and signed and stamped by the Inspector.

Mill certificates shall reference to MSS SP 44 together with the present specification and related data sheet.

The following documentation shall be submitted to the Inspector prior to the dispatch of the flanges:

- Mill test certificates signed by the Manufacturer to signify compliance with this specification.
This shall contain the following:
 - a. Manufacturer's name and trademark
 - b. Product serial number/Requisition No.
 - c. Flange Diameter and rating
 - d. Reference/Contract number
 - e. Heat number
 - f. Test number
 - g. Chemical analysis (ladle and product) including Carbon Equivalent
 - h. All mechanical test results
 - i. Heat treatment batch number.
 - Heat treatment chart signed by the Inspector, showing date of treatment and definition of scales
 - Sketch showing the location on the flange where the test specimen is removed from
 - Dimensional and NDT reports
 - Welding Procedure Qualification Records and Non-Destructive Testing's made on production welds (if any)
 - Hydrostatic testing certificate of compliance
 - Quality Control Plan
 - Compilation of concessions/deviations, if any, granted by the Company during flange production.

The number of originals and copies of above documentation shall be stated in the Project Particular Specification.

The original documentation shall be stamped and signed by the Inspector. The copies shall be stamped by the Inspector